Nickolas Simons

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Education

University of Utah — Salt Lake City, UT
Master of Entertainment Arts and Engineering, Engineering Track — *Aug. 2026 – May 2028*

University of Illinois at Urbana-Champaign — Urbana, IL
Bachelor of Science, Computer Science — *Aug. 2022 – May 2025*
Minor in Game Studies and Design — *GPA: 3.96, High Honors*

Experience

University of Utah, Division of Games — Salt Lake City, UT
Research Assistant - Digital Twins Research — *Aug. 2026 – Dec. 2026*

- Developed digital twins of department facilities using Unreal Engine and LiDAR scanners

University of Illinois, stu/dio — Urbana, IL
Programming Team Lead — *Aug. 2024 – Dec. 2025*

- Oversaw a team of 12 programmers across 8 projects, conducting code reviews and providing planning feedback across all projects
- Maintained Perforce infrastructure and integrated Horde for automated builds and Helix Swarm for code review
- Established coding standards and development workflows to improve code maintainability across projects

Gameplay Programmer — *Apr. 2024 – Dec. 2025*

- Designed and implemented gameplay features and tools for VR platforms using C++ and Blueprints, with an emphasis on accessibility, performance, and usability in classroom environments
- Optimized gameplay code and materials to meet VR performance targets using profiling tools like Unreal Insights and NVIDIA Nsight

Projects

Recycool | *UE5, C++, Blueprint, Perforce* — Sep. 2026

- Built an educational VR game for standalone Meta Quest in four weeks, teaching children how to properly recycle, incorporating street art and trick shot mechanics for greater user engagement
- Implemented an aim-assist system using C++ and Blueprint to make throwing and physics-based interactions more intuitive for VR users
- Worked in a team of 6, developing data-driven design interfaces and drafting technical documentation for more efficient on-boarding and team communication

VRchaeology | *UE5, C++, Blueprint, Perforce* — May 2025 – Dec. 2025

- Designed and implemented a quest system with a GUI editor using C++ and Blueprint, allowing designers to guide user actions through reusable scripting logic and contextual prompts
- Developed an interactive, tool-based VR excavation system using C++, simulating surface digging and stratigraphic layer analysis
- Implemented shared sublevel instances in UE5 to maintain continuity across multiple uses of a shared environment, ensuring consistent scene behavior and simplifying content team workflows

Multiplayer Action Game | *UE5, C++, Blueprint, ASIO, C#, GraphQL, MongoDB* — Nov. 2025 – Apr. 2026

- Built a replicated combat framework using C++ and Epic's Gameplay Ability System, implementing client-side prediction and correction for movement abilities
- Developed an online service using C#, MongoDB, and GraphQL to integrate with matchmaking, allowing queries for game state information prior to joining a session
- Created a system for generating procedural environments using Unreal Engine's Procedural Content Generation framework

Technical Skills

Languages: C++, C, C#, SQL, Python, UE Blueprint
Tools: Unreal Engine 5, Perforce, Git, Visual Studio, Unity 6, Docker, Horde, Unreal Insights, NVIDIA Nsight
Technologies: SDL3, CMake, ASIO (C++), Hot Chocolate (GraphQL)